

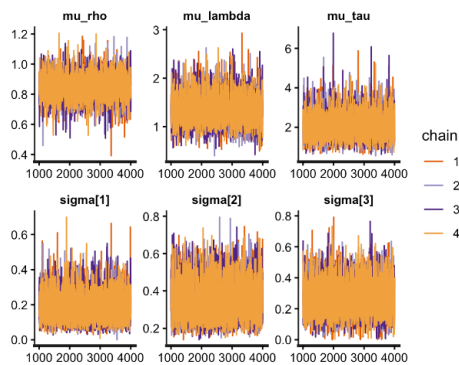
HW5

Problem 1

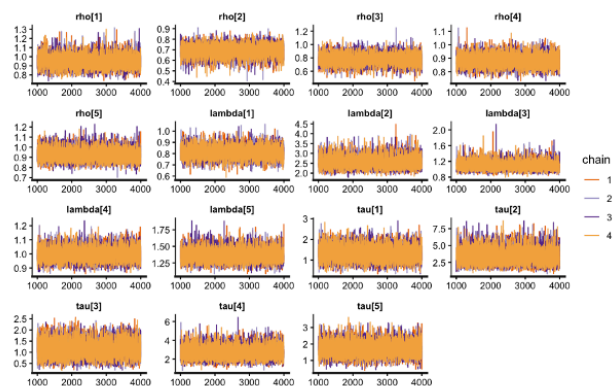
1.0. Convergence check

Convergence diagnostics indicated that the hierarchical Bayesian model converged successfully. Convergence was assessed using four MCMC chains and for both group-level parameters (mean and standard deviation of each parameter distribution) and individual-level parameters, the $\hat{R}$ values were almost 1.00. In addition, the trace plots of the four MCMC chains showed good mixing without noticeable divergence or non-convergence patterns.

Trace plots of group-level parameters

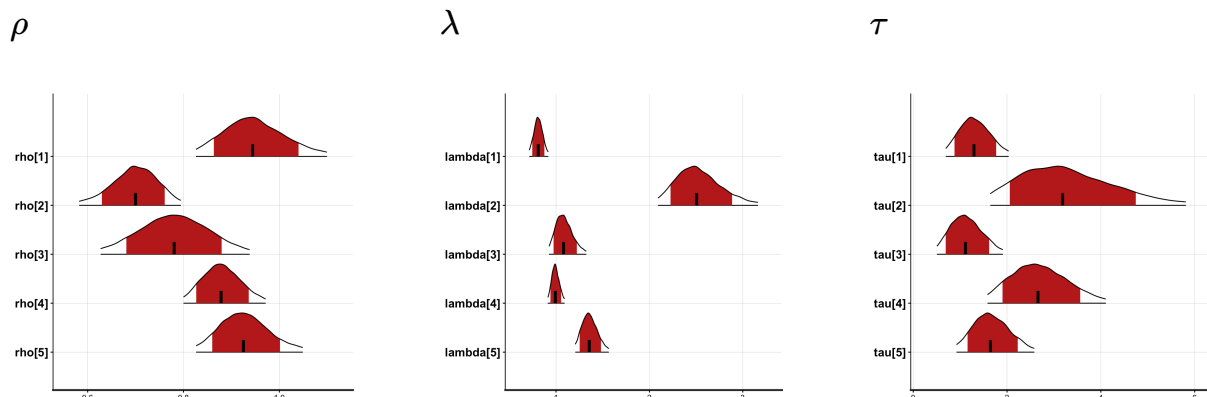


Trace plots of individual parameters



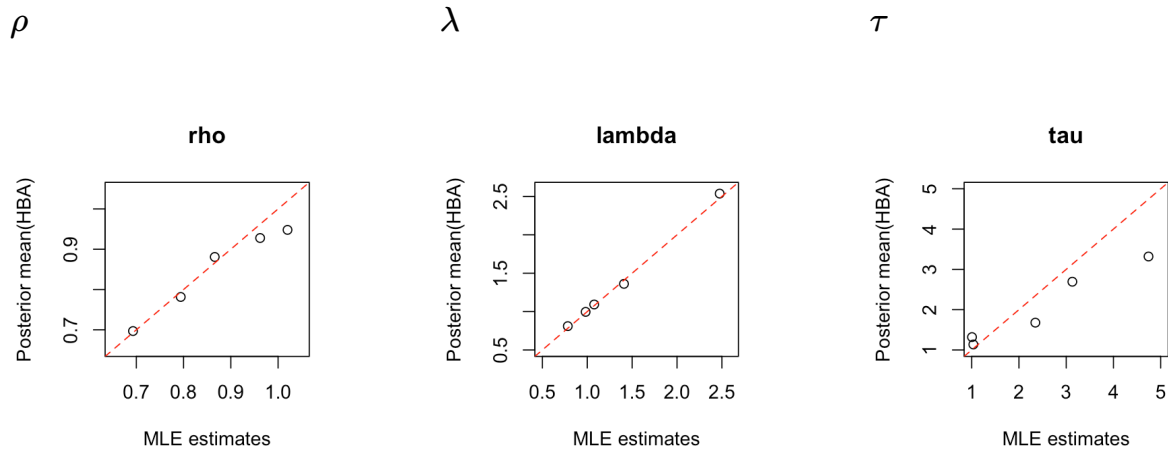
1.1. Posterior distributions of the five subjects and comparison of MLE estimates and posterior means

The posterior distributions of the individual parameters (rho, lambda and tau) are shown below.



1.2. Comparison of posterior means and MLE

Comparisons between the MLE estimates from HW2 and the posterior means obtained from the hierarchical Bayesian model are shown below. In the comparison plots, the red dotted line indicates the identity line ($x=y$). The posterior means for the rho (risk aversion) and tau (inverse temperature) parameters obtained from hierarchical Bayesian estimation appeared less dispersed across individuals than the corresponding MLE estimates. In particular, subjects with relatively high MLE estimates of rho and tau showed lower posterior mean estimates that moved closer to the group mean.



1.3 Reason for the difference between the two approaches

Unlike individual MLE fitting, the hierarchical Bayesian model assumes that individual parameters are drawn from a common group-level distribution. As a result, parameter estimation is influenced not only by each individual's data, but also by the overall group distribution. This produces a shrinkage effect, where extreme individual parameter estimates are pulled toward the group mean. As shown in 1.2, the posterior means from the hierarchical Bayesian model were generally less dispersed across individuals compared to the MLE estimates.

Problem 2

2.a.

From HW2, the summed BIC across subjects of two models are 651.863 (ra_noLA) and 580.435 (ra_prospect).

Approximately,

$$BF_{12} = \frac{p(D|M_1)}{p(D|M_2)} \approx \exp\left(\frac{BIC_2 - BIC_1}{2}\right)$$

Using this, bayes factor of **ra_prospect (M_1) over ra_noLA (M_2)** is approximately

$$BF_{\text{prospect,noLA}} \approx \exp\left(\frac{651.863 - 580.435}{2}\right) \approx 3.24 * 10^{15}$$

2.b

According to Jeffery's evidence categories, $BF_{12} > 100$ indicates extreme evidence for M_1 . Thus, the result shows extreme evidence for ra_prospect, over ra_noLA

2.c

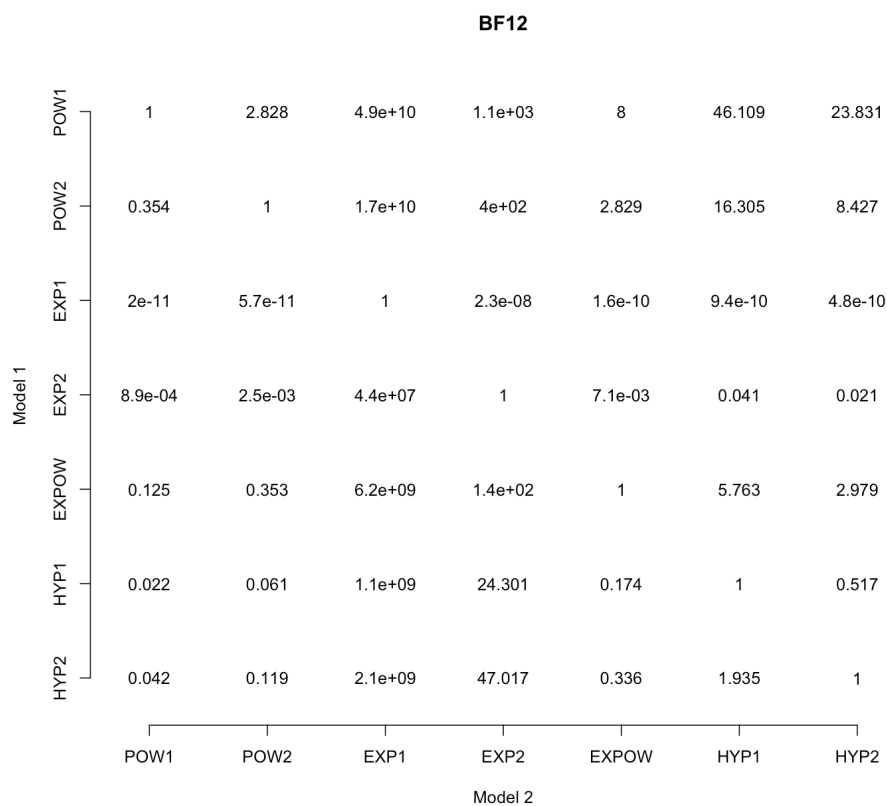
$$\frac{p(M_1|D)}{p(M_2|D)} = \frac{p(D|M_1)p(M_1)}{p(D|M_2)p(M_2)} = BF_{12} * \frac{p(M_1)}{p(M_2)} \approx 4.86 * 10^{15}$$

2.d

$$\begin{aligned} p(M_2|D) &= 1 - p(M_1|D) \\ \therefore \frac{p(M_1|D)}{1 - p(M_1|D)} &\approx 4.86 * 10^{15} \\ \therefore p(M_1|D) &\approx \frac{4.86 * 10^{15}}{1 + 4.86 * 10^{15}} = 0.999... \end{aligned}$$

2.e

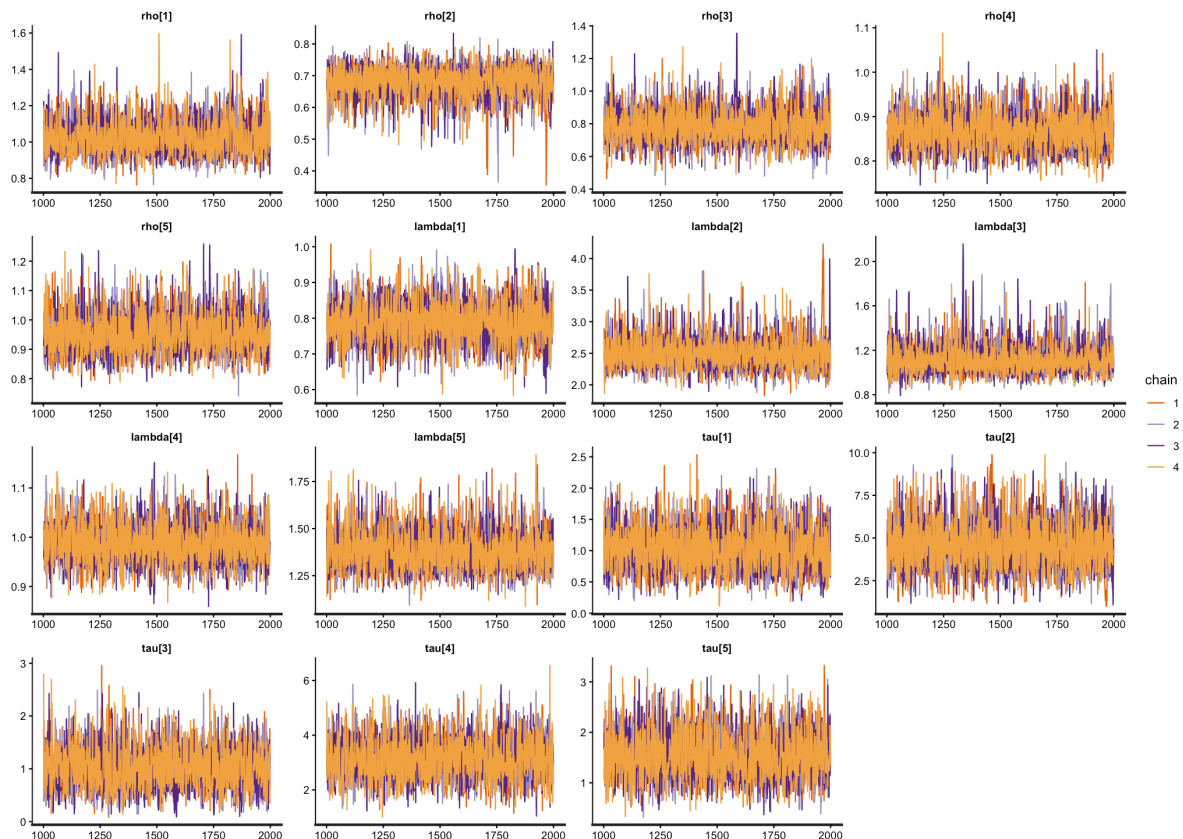
The figure below shows approximate bayes factor of model 1 (y-axis) over model 2 (x-axis)



Problem 3

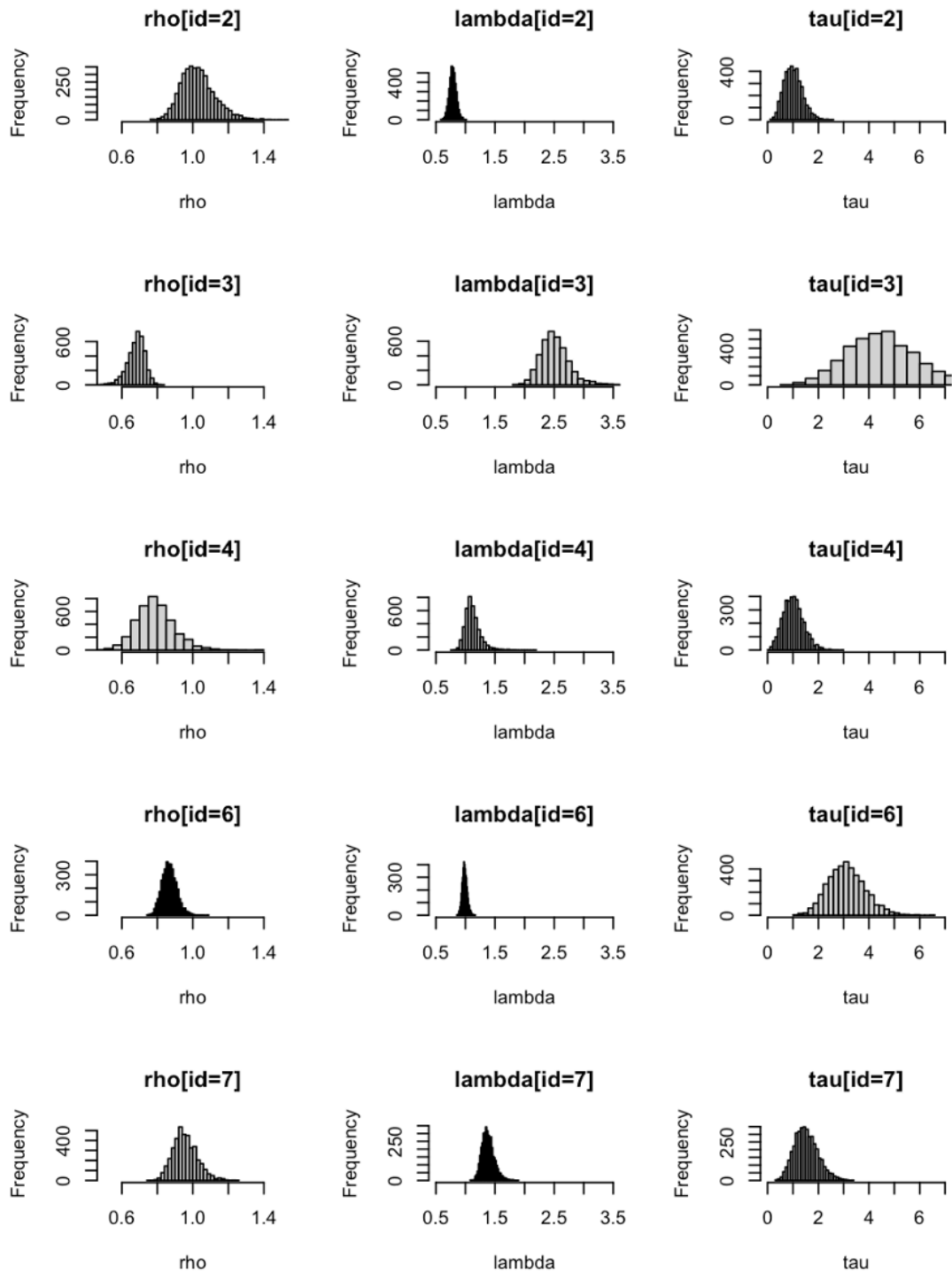
3.0. Convergence check

Convergence diagnostics indicated that the hierarchical Bayesian model converged successfully. Convergence was assessed using four MCMC chains. For all individual parameters, $\hat{R}$ values were almost 1.00. The trace plots of the four MCMC chains showed good mixing without noticeable divergence or non-convergence patterns.



3.1. Posterior distribution of 5 subjects

The figure below shows posterior distributions estimated from the MCMC samples for each parameter of each individuals.



3.2 Posterior means of 5 subjects

For each parameters, the posterior mean of each subject are suggested as the table below.

ID	ρ	λ	τ
2	1.029	0.787	1.011
3	0.68	2.516	4.586
4	0.788	1.122	1.019
6	0.868	0.988	3.119
7	0.957	1.383	1.526